MCU

文件: MCU.kicad_sch

POWER_SHARED

文件: power.kicad_sch

FOC_M1

文件: FOC_M1.kicad_sch

FOC_M2

文件: FOC_M2.kicad_sch

Dual independent motor framework
Common input supply + GD32F505VxTx controller
Review protection thresholds and ratings before energizing.

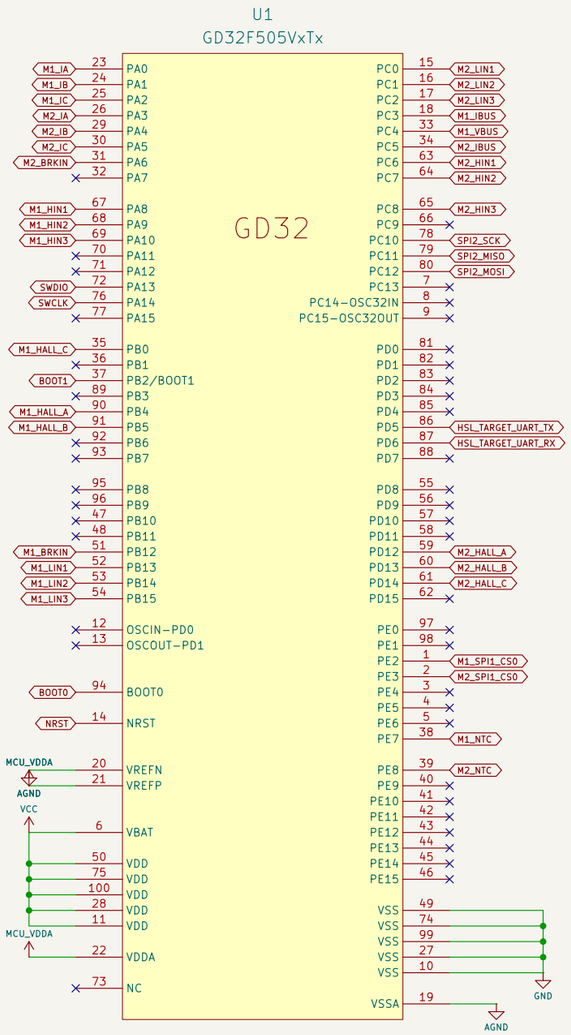
HSL_DEBUGGER

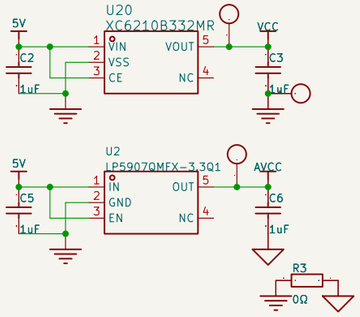
文件: HSLink.kicad_sch

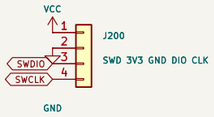
Sheet: /
File: GD32F5_FOC.kicad_sch

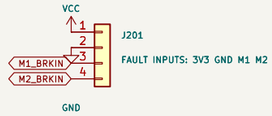
Title:

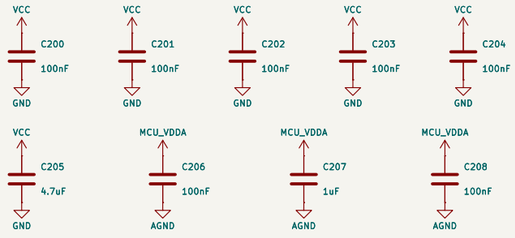
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6-87-gae6d8f40c4		Id: 1/6





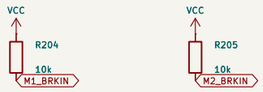










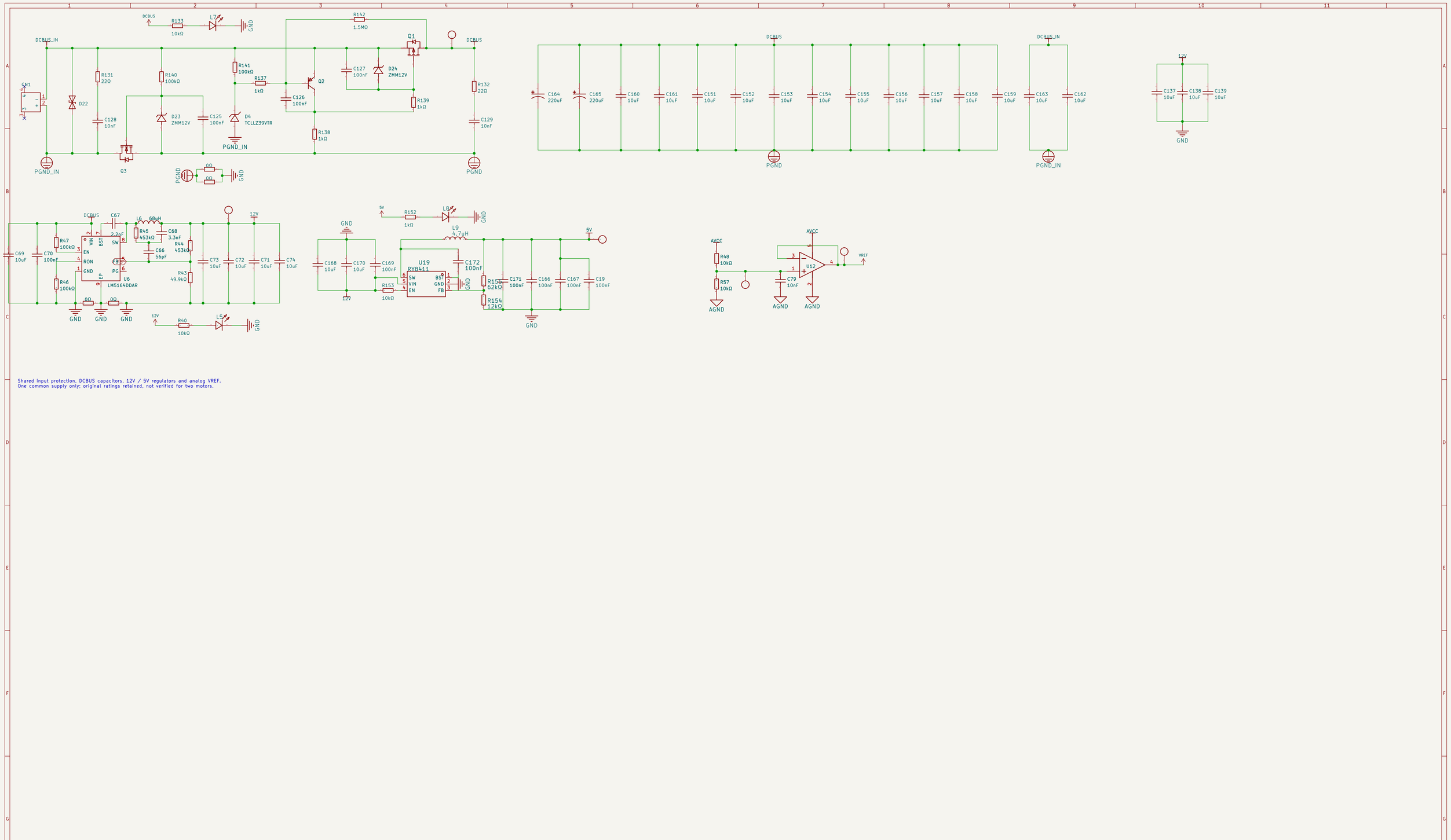


Onboard debugger: HSLink page; R8901..R8905 are default-open.
J200 external probe: never use simultaneously with onboard probe.
USART1 bridge PD5/TX, PD6/RX; target must be powered before enabling links.

MCU_VDDA = VCC through R203 (OR), bypass C206..C208 to AGND.
AVCC LDO remains for external analog amplifiers only.
J201: 1=3V3, 2=GND, 3=M1_BRKIN, 4=M2_BRKIN.
FAULT inputs use pullups; external open-drain HEALTH holds LOW only while healthy.
Configure each timer break polarity HIGH; unplugged defaults to brake. No autonomous trip comparator fitted.
Current limits/thresholds and external fault source remain to be designed.

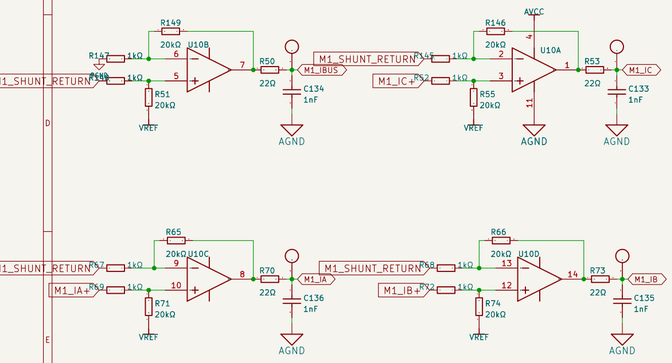
M1 TIMER0: PA8/9/10 AF1; PB13/14 AF2, PB15 AF1. BRKIN PB12 AF4.
M2 TIMER7: PC6/7 AF3, PC8 AF2; PC0/1/2 AF0. BRKIN PA6 AF4.
M1 currents PA0/1/2; M2 currents PA3/4/5 (ADC0/ADC1).
IBUS: PC3 / PC5; VBUS PC4; NTC PE7 / PE8 (ADC2).
Hall M1 PB4/5/0 TIMER2; M2 PD12/13/14 TIMER3. Release PB4 JTAG, keep SWD.
Shared SPI2 PC10/11/12 AF2; GPIO chip selects PE2 / PE3.
VA/VB/VC/COM and M2_VBUS stay on power-sheet testpoints, not MCU ADC.

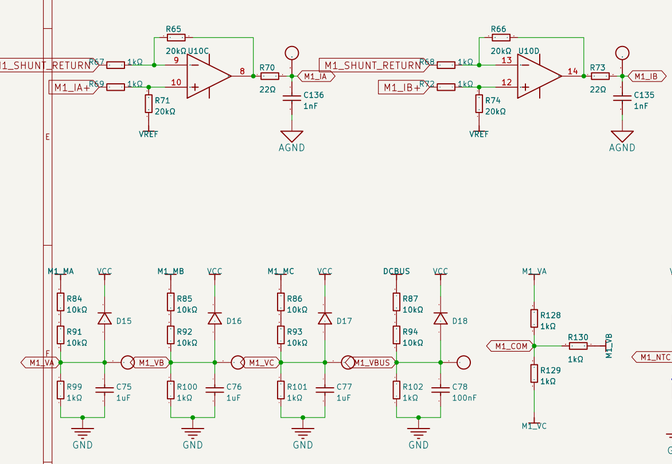
Sheet: /MCU/
File: MCU.kicad_sch
Title:
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.6-87-gae6d8f40c4 | Id: 2/6

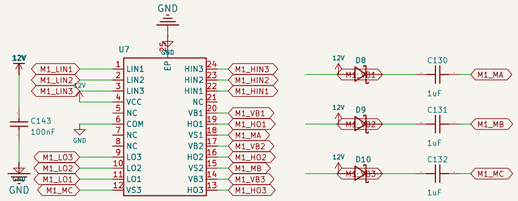


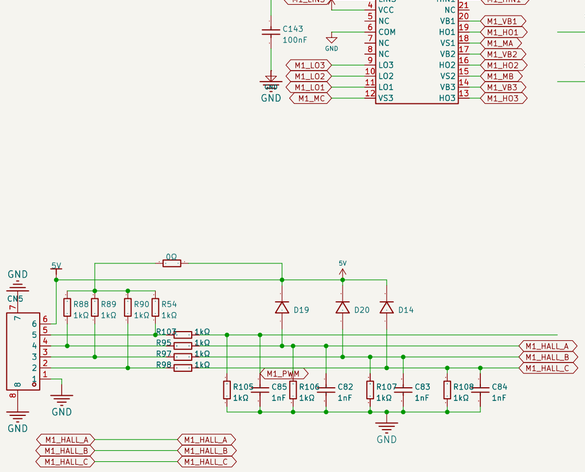
Shared input protection, DCBUS capacitors, 12V / 5V regulators and analog VREF.
One common supply only; original ratings retained, not verified for two motors.

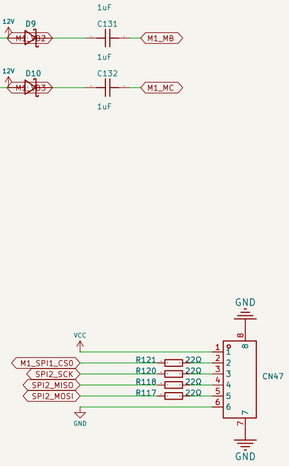
Sheet: /POWER_SHARED/		
File: power.kicad_sch		
Title:		
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.6-87-gae6d8f40c4		Id: 3/6

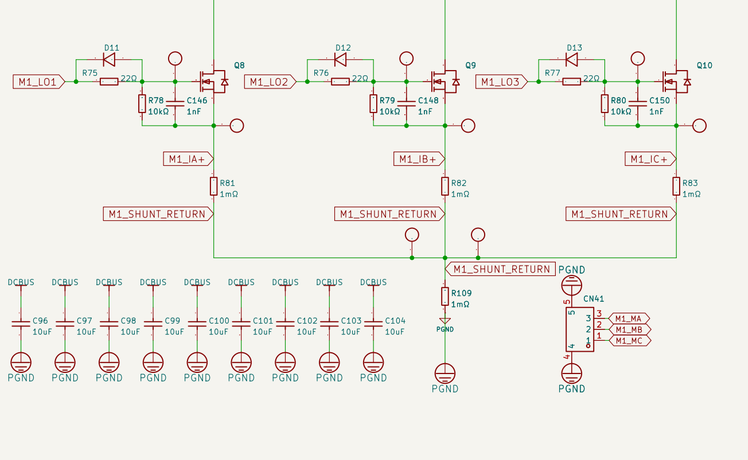










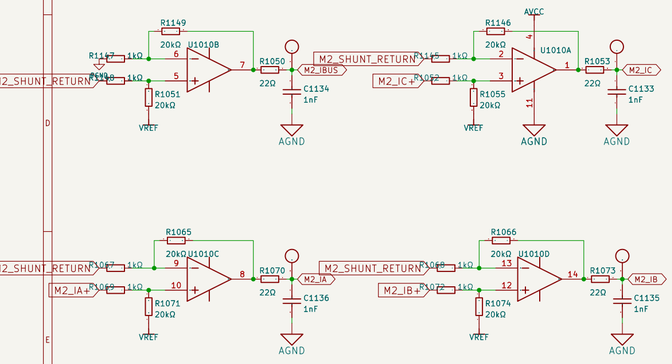


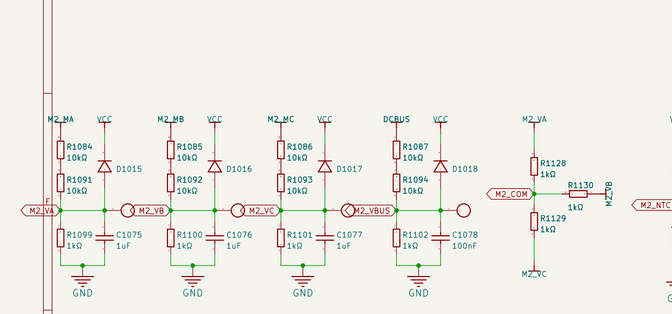


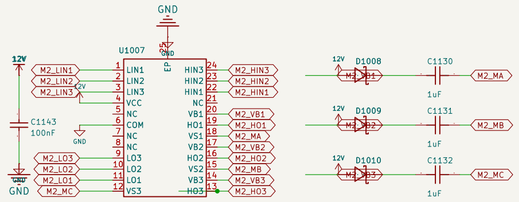
MOTOR 1: Independent gate drive, six MOSFETs, phase shunts and bus shunt.
All motor signals M1_*. Common DCBUS / 12V / 5V / VCC / AVCC / VREF.
PGND is common AFTER the dedicated bus shunt; phase-shunt return remains local.
Reference values and footprints require engineering review; no rating verification.

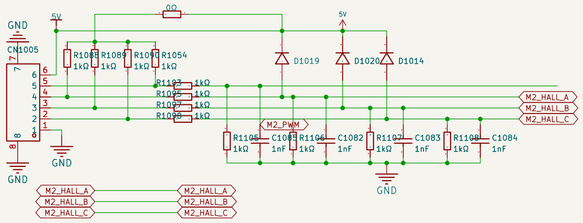
VA/VB/VC, COM, IBUS, NTC and auxiliary PWM are retained reference interfaces.
Only nets explicitly connected on MCU sheet are MCU functions.
FD6288Q has no dedicated EN pin; hardware gate-input pulldowns are provided.
BRKIN requires external fault generation; software must initialize PWM disabled.

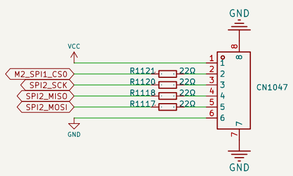
Sheet: /FOC_M1/	
File: FOC_M1.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.6-87-gae6d8f40c4	Id: 4/6

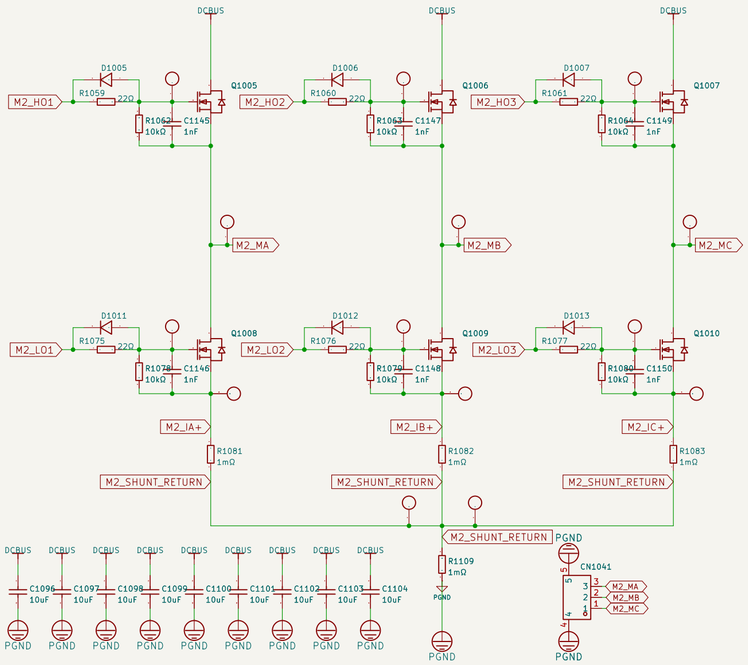


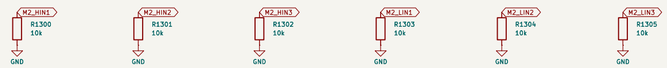












MOTOR 2: Independent gate drive, six MOSFETs, phase shunts and bus shunt.
All motor signals M2_*. Common DCBUS / 12V / 5V / VCC / AVCC / VREF.
PGND is common AFTER the dedicated bus shunt; phase-shunt return remains local.
Reference values and footprints require engineering review; no rating verification.

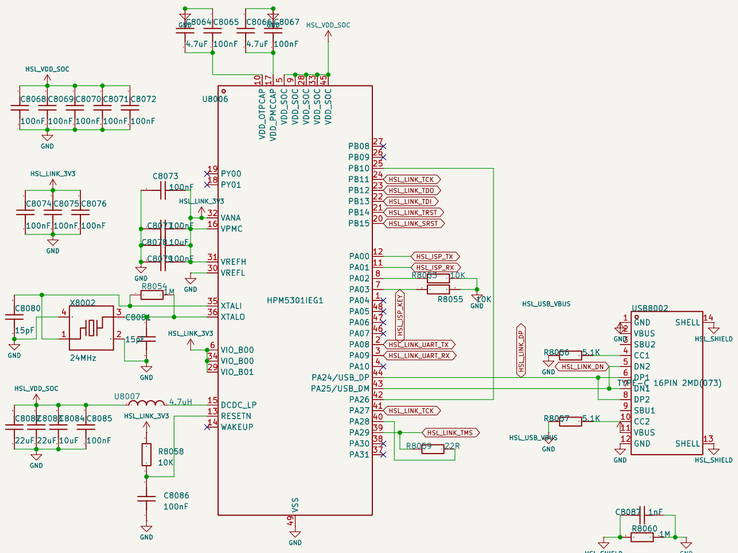
VA/VB/VC, COM, IBUS, NTC and auxiliary PWM are retained reference interfaces.
Only nets explicitly connected on MCU sheet are MCU functions.
FD6288Q has no dedicated EN pin; hardware gate-input pulldowns are provided.
BRKIN requires external fault generation; software must initialize PWM disabled.

Sheet: /FOC_M2/
File: FOC_M2.kicad_sch

Title:

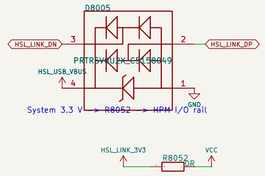
Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.6-87-gae6d8f40c4		Id: 5/6

HSLINK / HPM5301 — SHARED SYSTEM POWER

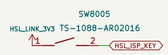


HPM5301 + 24 MHz clock + on-chip VDD_SOC buck

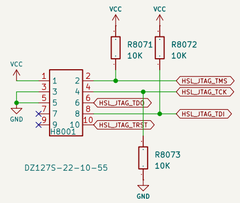
USB data / ESD: VBUS does not power board



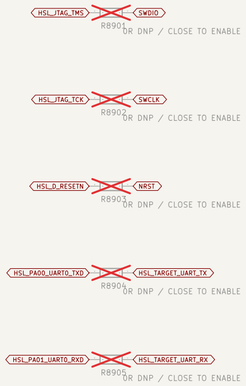
ISP boot button



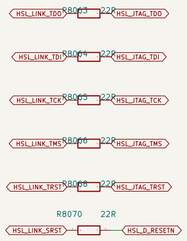
Target reference header



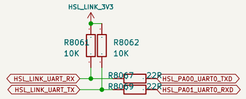
Target disconnect links (default DNP)



JTAG series termination



UART series termination



POWER: VCC = existing system 3.3 V (U20 XC6210B332MR).
R8052 OR supplies HSL_LINK_3V3: HPM on-chip VDD_SOC buck retained.
USB VBUS only reaches USB connector / ESD; no board-supply path.
U20 load / thermal margin requires measurement; HPM load is unknown.

R8901..R8905 remain DEFAULT DNP / OPEN. Fit only with target powered.
Never drive target simultaneously from onboard debugger and J200 probe.
H8001 pin 1 is target VCC reference; no separate power injection.
No validated ADC voltage sensing or firmware power-off tri-state.

Reference GPIO mapping retained; firmware compatibility NOT verified.
USB footprint and firmware require validation before PCB fabrication.

UART nets name target direction: TX = GD32 PD5 -> probe RX;
RX = GD32 PD6 <- probe TX.

Sheet: /HSL_DEBUGGER/
File: HSLink.kicad_sch
Title:
Size: A2 | Date: | Rev:
KiCad E.D.A. 10.0.6-87-gae6d8f40c4 | Id: 6/6